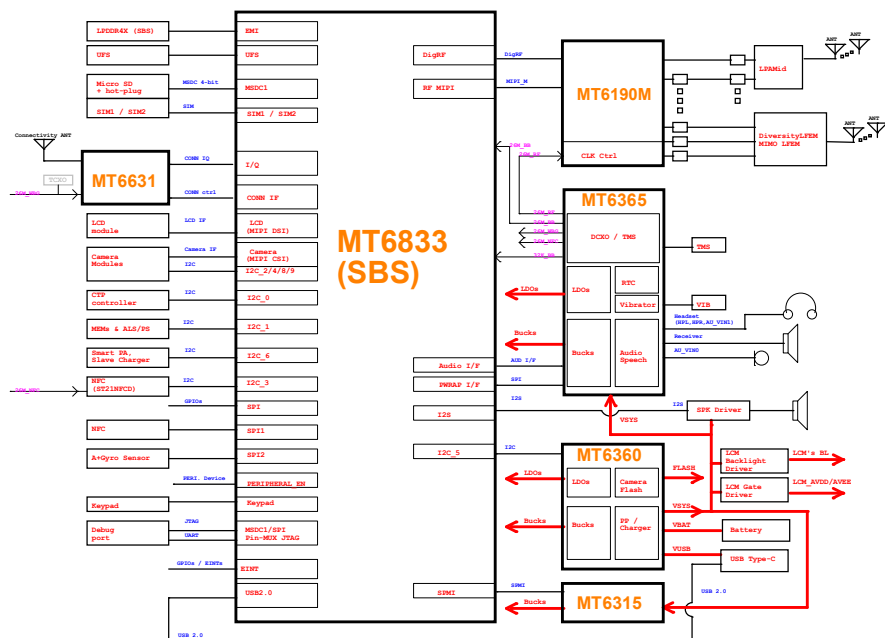
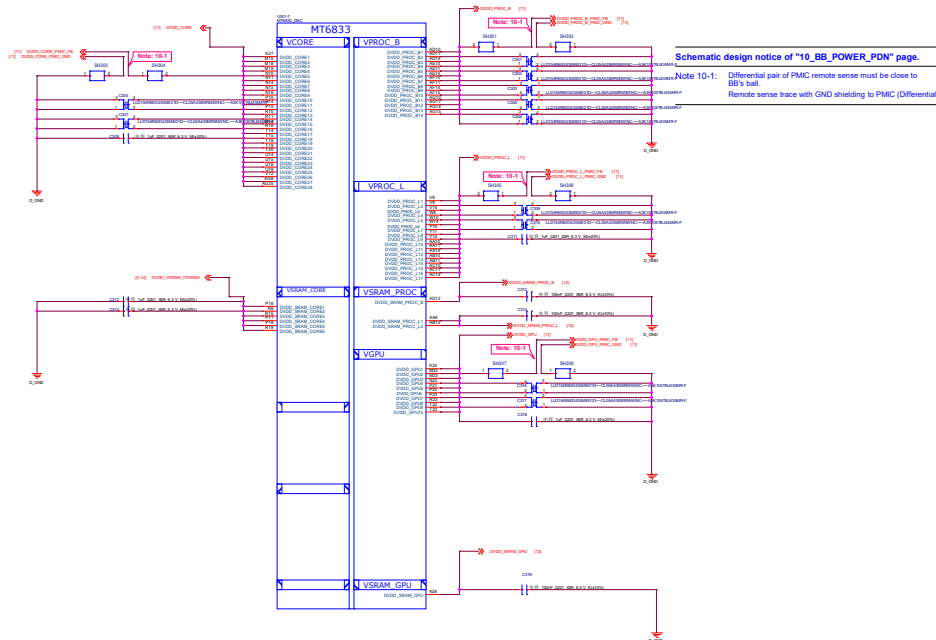




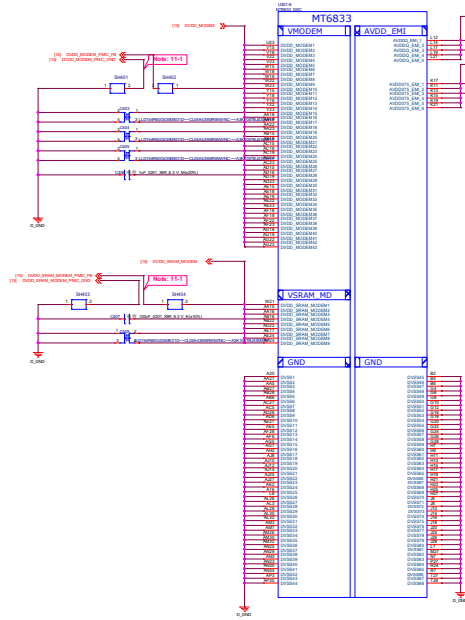
华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		Block Diagram	
Size	Project	<Doc>	Rev
D			V1
Date:	Monday, October 18, 2021	Sheet 1 of 50	

Date	Version	Page	Change notice
2020.06.xx	Pre-Released		
2020.07.xx	V01		
2020.08.14	V02	13_BB_1	Disconnect Dummy pins A1,A34,AP1,AP34 to GND
		29_POWER_EXT	Change C2923 cap from 6.3V to 10V
		20_POWER_MT6365_Buck	[PDN] Change Caps of DVDD_GPU - C2039/C2038 to NC
		25_POWER_MT6315_VMD	[PDN] Change Caps of DVDD_MODEM - C2502/C2534/C2523/C2522 to NC
2020.08.26	V03	File name	Modify the file name

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title SM6350_Control		
Size D	Project <Doc>	Rev V1
Date: Monday, October 18, 2021	Sheet 2	of 50

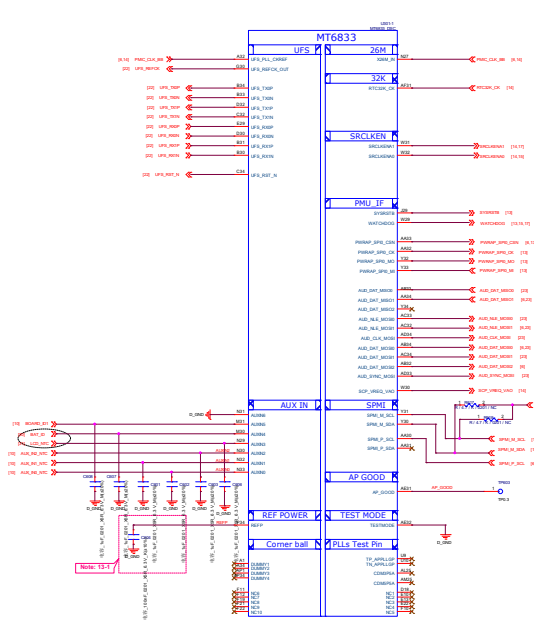


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 03.BB_POWER_PDN1			
Size D	Project	<Doc>	Rev v1
Date: Monday, October 18, 2021	Sheet 3 of 23		



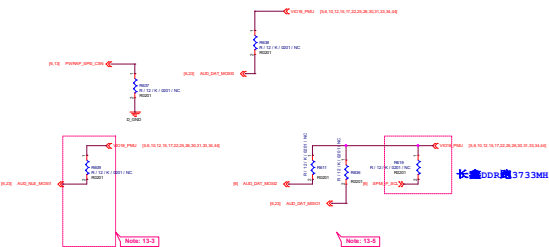
Schematic design notice of "11_BB_POWER_PDN" page.
 Note 11-1: Differential pair of PMC remote sense must be close to SIO's ball.
 Remote sense trace with GND shielding to PMC (Differential)

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 04.BB_POWER_PDN2			
Size D	Project	<Doc>	
Date: Monday, October 18, 2021		Sheet 4 of 23	Rev v1



Note 13-6: "AUD_DAT_MISO1" is trapping pin to select VEMC Voltage.

AUD_DAT_MISO1	VEMC Voltage
L (Default)	VEMC=3.0V for VCC of UFS2.1 device
H (by external PU)	VEMC=2.5V for VCC of UFS3.0 device



Schematic design notice:

Note 13-1: The load cap. have to be placed as close to REFP ball as possible.

Note 13-2: "PWRAP_SPI0_CSN" and "AUD_DAT_MISO0" pin features in trapping pin to enable JTAG.

PWRAP_SPI0_CSN	AUD_DAT_MISO0	AP_JTAG	ID_JTAG
H (Default)	L (Default)	N/A	N/A
(by external PU)	(by external PU)	SPINENT8	SPINENT8
L (by external PD)	L	SPI0-ENT8	SPINENT8
L (by external PU)	H (by external PU)	MISO0_CLK MISO0_DAT0 MISO0_DAT1 MISO0_DAT2	N/A

Note 13-3: "AUD_NLE_MOSI1" features in trapping pin to enable JTAG over USB 2.0 port.

AUD_NLE_MOSI1	USB_JTAG
L (Default)	Disable serial JTAG over USB
H (by external PU)	DPIDM output JTAG

Note 13-4: "AUD_SYNC_MOSI" and "AUD_NLE_MOSI0" pin features in trapping pin to booting (eMMC/UFS/SPI NOR).

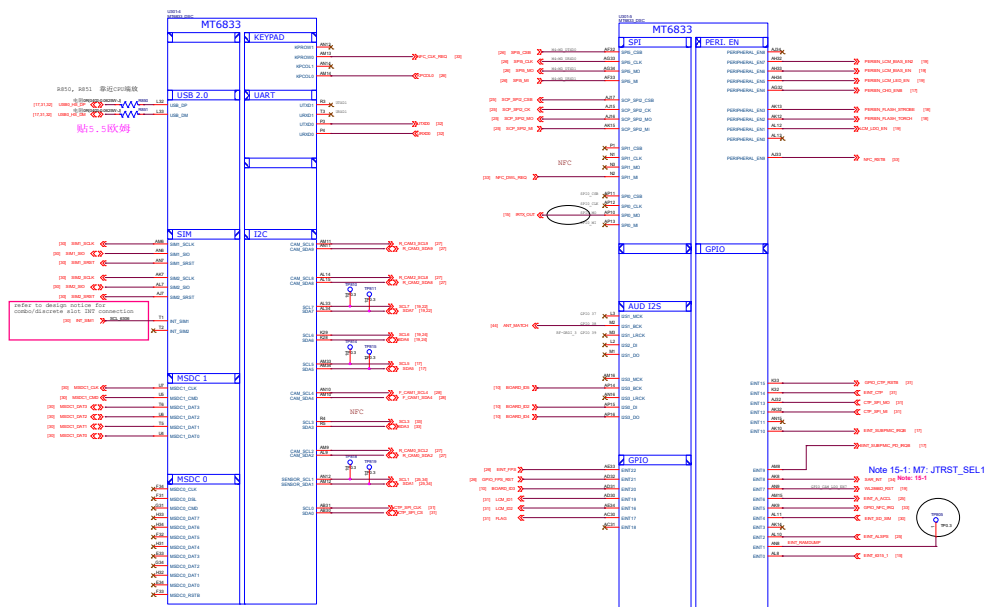
AUD_SYNC_MOSI	AUD_NLE_MOSI0	Storage Booting
L (Default)	L (Default)	Only UFS boot
H (by external PU)	H	Only eMMC boot
H (by external PU)	L	Reserved
H (by external PU)	H (by external PU)	Reserved

Note 13-5: "AUD_DAT_MOSI1" and "AUD_DAT_MOSI2" pin features in trapping pin to enable DDR.

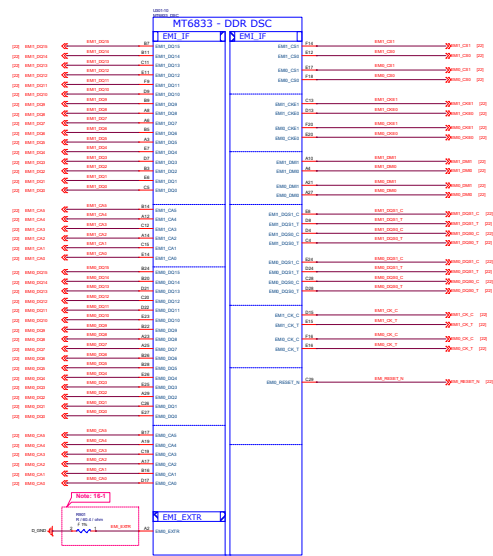
AUD_DAT_MOSI1	AUD_DAT_MOSI2	SPI0_P_SCL	DDR
L (Default)	L (Default)	L (Default)	LP4xDDR4-2133MHz
H (by external PU)	H (by external PU)	L (Default)	LP4xDDR4-2133MHz
H (by external PU)	L	L (Default)	Reserved
H (by external PU)	H	L (Default)	Reserved

Note 1: SPI0_P_SCL for DDR 4200Mbps/2133Mbps Co-load, 1.2V = 1200Mbps.
*H (by external PU) = 2133Mbps

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Title 06.BB_1		
Size D	Project <Doc>	Rev v1
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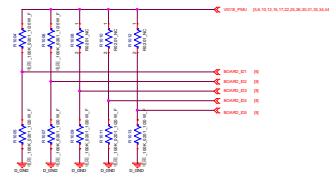
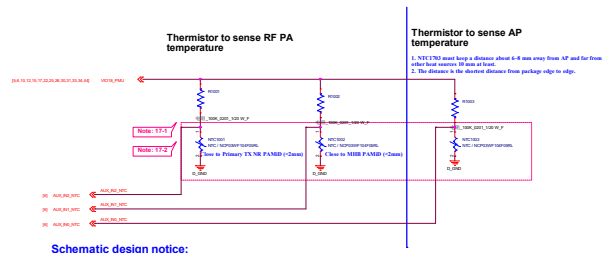
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		08.BB_3	
Size	Project	<Doc>	Rev
D			v1
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Schematic design notice of "16_BB_4 Interface" page:

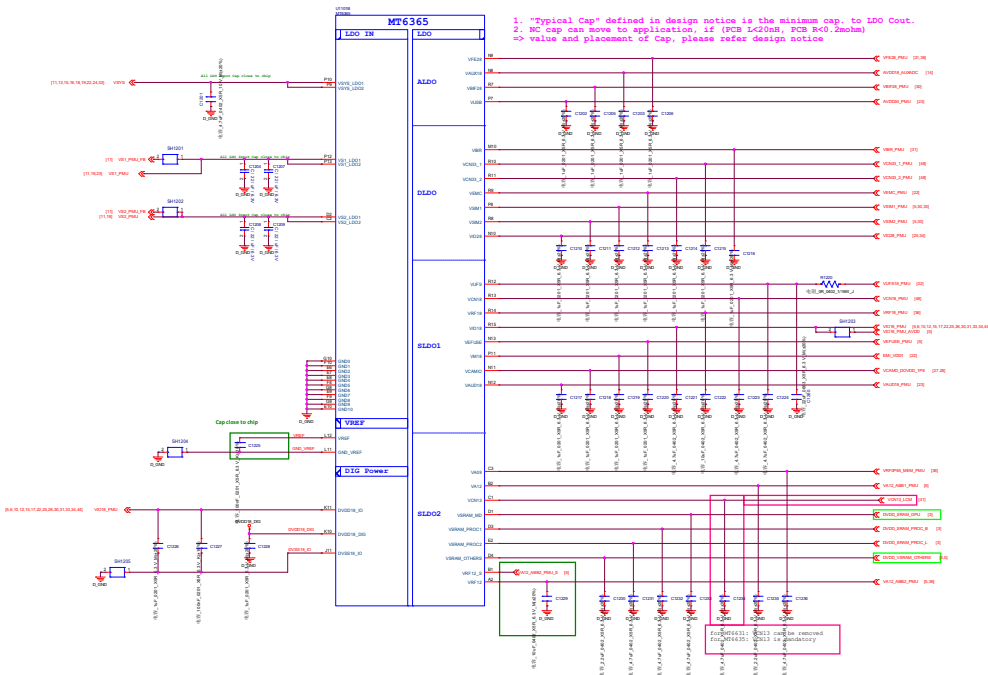
Note 16-1: R4001 please select 60.4 ohm (1%) resistor

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		09.BB_4	
Size	Project	<Doc>	Rev
D			v1
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华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		10.BB_AUXADC_Thermal/BOARD_ID	
Size	Project	<Doc>	Rev
D			V1
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Note: Refer to Platform SoC Design Notice for PMIC part number



1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.
2. MC cap can move to application, if (PCB 1x20mm, PCB R40.2mm) => value and placement of Cap, please refer design notice

Circuit Type	LDO Name	Output Voltage (V)	Boost (Default) (V)	Current (mA)	Expected use
ALDO	VFE2B	2.80	OFF (2.8)	200	RFEE
	VU1B	3.37	OFF (3.37)	200	U1B
	VALU1B	1.84	OFF (1.84)	50	ALUADC
	VIO22	2.24	ON (2.24)	25	DCKO
VRTC	VRTC	2.80	OFF (2.8)	50	Battery Interface
	DIVDD1B_D10	1.80	ON (1.8)	10	PMIC Digital

Circuit Type	LDO Name	Output Voltage (V)	Boost (Default) (V)	Current (mA)	Expected use
DDO	VSM1	1.7/1.8/1.9/2.0	OFF (1.8)	200	SIM
	VSM2	2.7/3.1/3.3	OFF (3.3)	300	SIM
	VCM33_1	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VCM33_2	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VEMC	2.55/2.7/3.0/3.3	ON (3.0)	800	eMMC / UFS
	VIO2B	2.8/2.9/3.0	OFF (2.8)	200	IO & Sensor
	VIBR	2.7/2.8/3.0/3.3	OFF (2.8)	200	Vibrator

Circuit Type	LDO Name	Output Voltage (V)	Boost (Default) (V)	Current (mA)	Expected use
SDIO1	VDD1B	1.80	ON (1.8)	800	SDIO
	VDD1B1	1.80	ON (1.8)	800	SDIO
	VDD1B2	1.80	ON (1.8)	800	SDIO
	VDD1B3	1.80	ON (1.8)	800	SDIO
	VDD1B4	1.80	ON (1.8)	800	SDIO
	VDD1B5	1.80	ON (1.8)	800	SDIO
	VDD1B6	1.80	ON (1.8)	800	SDIO

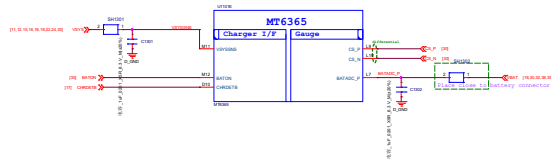
Circuit Type	LDO Name	Output Voltage (V)	Boost (Default) (V)	Current (mA)	Expected use
SDIO2	VDD2B	1.80	ON (1.8)	800	SDIO
	VDD2B1	1.80	ON (1.8)	800	SDIO
	VDD2B2	1.80	ON (1.8)	800	SDIO
	VDD2B3	1.80	ON (1.8)	800	SDIO
	VDD2B4	1.80	ON (1.8)	800	SDIO
	VDD2B5	1.80	ON (1.8)	800	SDIO
	VDD2B6	1.80	ON (1.8)	800	SDIO

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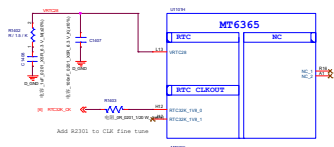
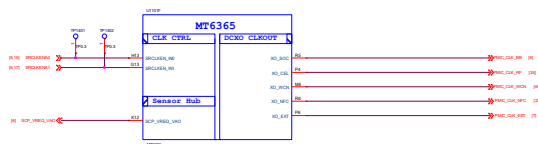
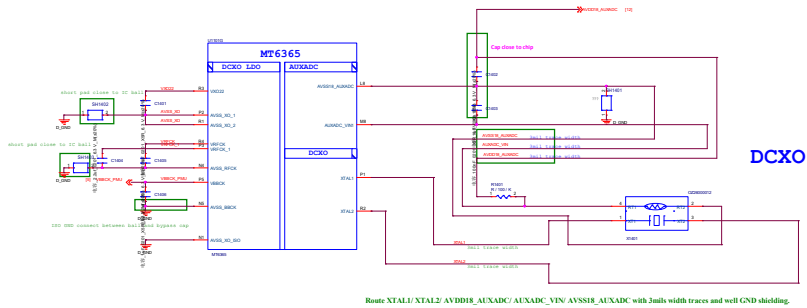
Title 12.POWER_MT6365_LDO

Size D Project <Doc> Rev v1

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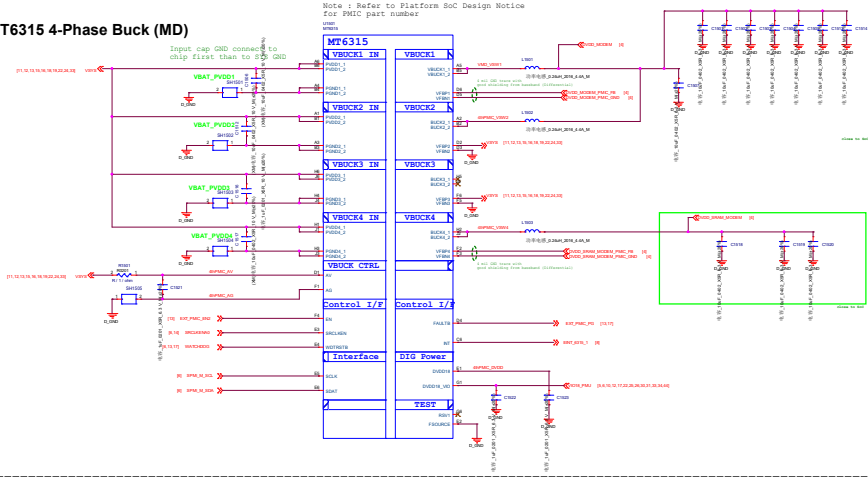


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 13.POWER_MT6365_General			
Size D	Project <Doc>	Rev v1	
Date: Thursday, November 11, 2010			
		Sheet 13 of 23	



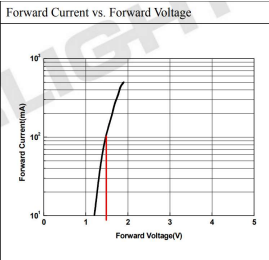
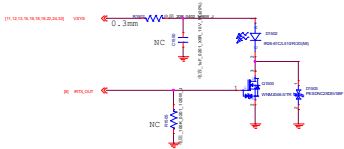
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		14.POWER_MT6365_Clock	
Size D	Project <Doc>	Rev v1	
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MT6315 4-Phase Buck (MD)



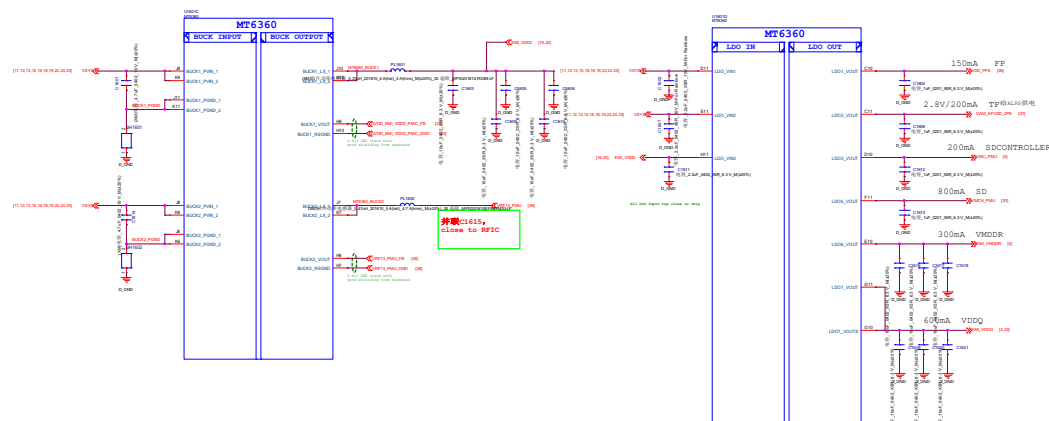
MT6315NP/B	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAx} (mA)	Expected use
BUCK1+2	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000*2	DVDD_MODEM
BUCK3	0.40 ~ 1.19375 (6.25mV/step)	ON (0.75)	5000	---
BUCK4	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000	DVDD_SRAM_MODEM

IR LED



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 15.POWER_MT6315_VMD/IR_LED			
Size D	Project <Doc>	Rev v1	
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Note : Refer to Platform SoC Design Notice for PMIC part number



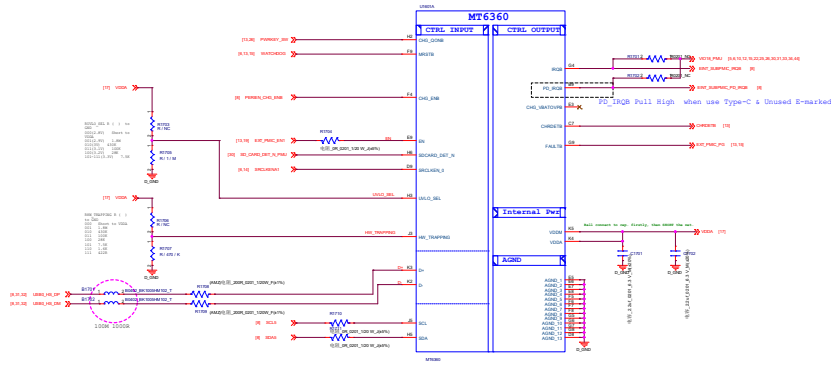
LDO1 VFP
LDO2 VTP
LDO3 VMC
LDO5 VMC
LDO6 EMI_VDDDR
LDO7 EMI_VDDQ

Circuit Type	Name	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
Buck	BUCK1	0.3 ~ 1.3 (5mV/step)	ON (1.125)	3000	EMI_VDD2
	BUCK2	0.3 ~ 1.3 (5mV/step)	OFF (1.3)	3000	VRF13_PMU

Circuit Type	Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
LDO	LDO1	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	150	Fingerprint
	LDO2	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	200	Touch Panel
	LDO3	1.8/2.9/2.8/3.0/3.3	OFF (3.0)	200	SD Card
	LDO5	2.9/3.0/3.3	OFF (3.0)	800	SD Card
	LDO6	0.75	ON (0.75)	300	EMI_VMDDR
	LDO7	0.6	ON (0.6)	600	EMI_VDDQ

华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title 16.POWER_MT6360_BUCK_LDO*			
Size D	Project <Doc>	Rev V1	
Date: Monday, October 18, 2021	Sheet 16 of 23		

Note 1: Refer to Platform SoC Design Notice for PMIC part number



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 17.POWER_MT6360_General			
Size	Project	<Doc>	Rev
D			v1
Date:	Monday, October 18, 2021	Sheet	17 of 23

The schematic diagram illustrates the internal architecture and external connections of the MT6360 power management IC. The IC is divided into four main functional blocks: CHG (Charging), PP+VDD (Power Plane and VDD), RGB (RGB LEDs), and FLSD (Flash/EEPROM).

CHG (Charging) Block: This block handles USB input and charging. It includes a USB input (USB_IN) and a USB-to-GND short circuit. The USB-to-VDD connection is shown with a USB-to-VDD pin and a USB-to-GND pin. The USB-to-VDD connection is labeled "Short to GND when Unused E-marked".

PP+VDD (Power Plane and VDD) Block: This block manages the power plane and VDD. It includes a USB-to-VDD connection and a USB-to-GND connection. The USB-to-VDD connection is labeled "Short to GND when Unused E-marked".

RGB (RGB LEDs) Block: This block controls the RGB LEDs. It includes an RGB_LED pin and an RGB_LED pin. The RGB_LED pin is connected to the RGB_LED pin of the IC.

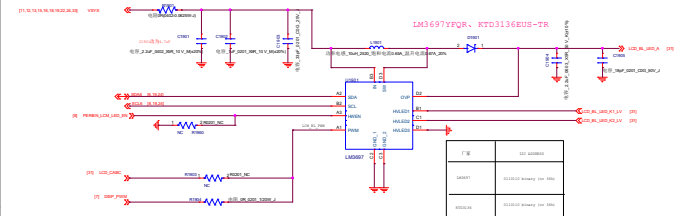
FLSD (Flash/EEPROM) Block: This block manages the Flash/EEPROM. It includes a FLSD pin and a FLSD pin. The FLSD pin is connected to the FLSD pin of the IC.

The diagram also shows various capacitors, resistors, and other components used in the power management circuit. The components are labeled with their values and part numbers. The diagram is a detailed representation of the MT6360 power management IC and its connections.

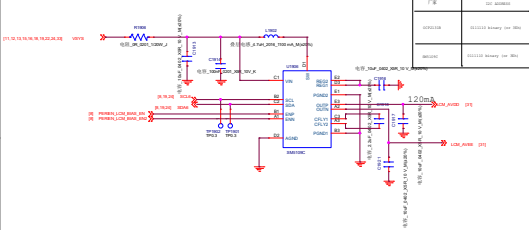
Note 28-1: For better ESD & surge performance we need choose suitable device for system protection. Please refer to the latest version of [Surge device selection guide] provided by MTK.

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 18.POWER_MT6360_Charger		
Size D	Project <Doc>	Rev v1
Date: Friday, December 24, 2021		
Sheet 18 of 23		

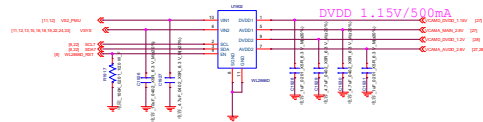
LCM Backlight LED Driver



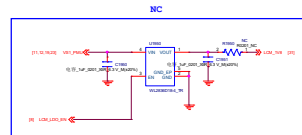
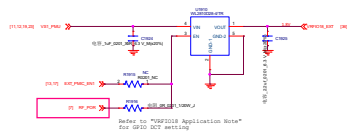
LCM Bias



U1	U2
MAX4422	MAX4422
MAX4422	MAX4422
MAX4422	MAX4422



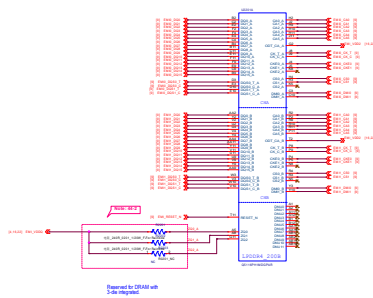
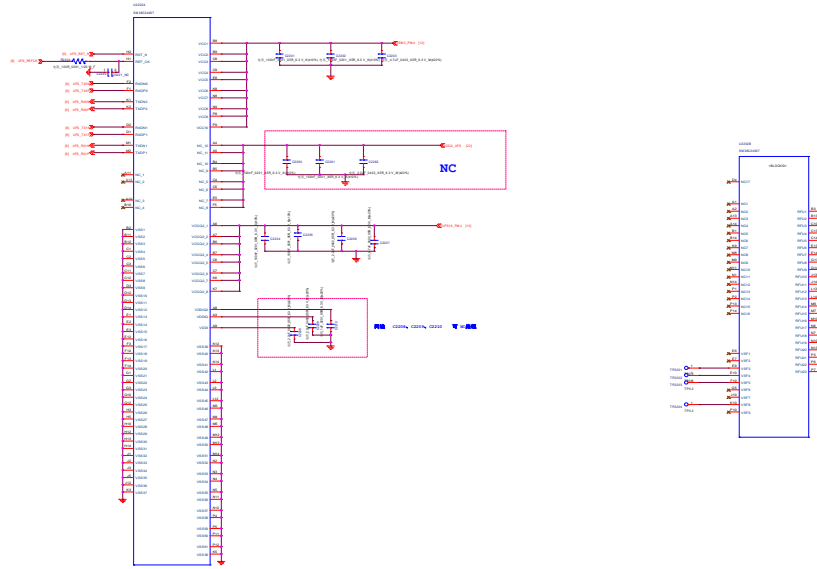
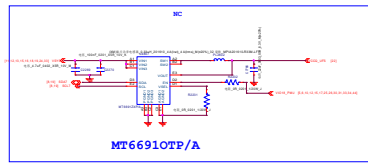
External VIO18_LDO: option for power leakage from UART port on product line



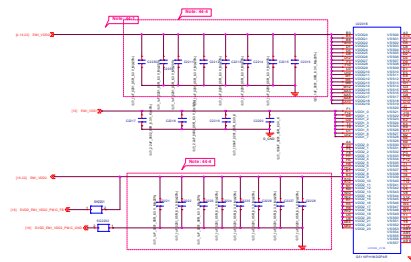
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		19.POWER_EXT	
Size	Project	<Doc>	Rev
D			v1
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华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title PM7250B_CHARGE/FG			
Size	Project	Rev	
E	<Doc>	v1	
Date:	Monday, October 18, 2021	Sheet	20 of 23

华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		PM7250B_MISC/GPIO/GND	
Size	Project	<Doc>	Rev
D			v1
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discrete LP4X
16Gb LPDDR4X

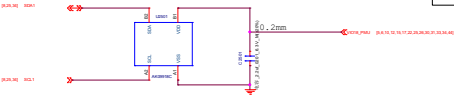


Schematic design notice of "44_Memory_4MMC_LPDDR4X"	
Note 44-1:	Please refer to power supply related page select: VDRAM2 / VDRAM output voltage property for LPDDR4X.
Note 44-2:	DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ.
Note 44-3:	Please refer to eMMC vendor's datasheet or MTK common design notice to get the recommendation typical cap. value for VDDQ/VDDQ2/VDDQ3 power domains of eMMC.
Note 44-4:	VDDQ VDDQ2 decoupling cap. (classed to DRAM bank). For other cap for PMIC >10uF, at PMIC page, please also refer to PMIC and layout guide for placement.

华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title 01_Block_Diagram		
Size E	Project J19_MB_V1	Rev v1
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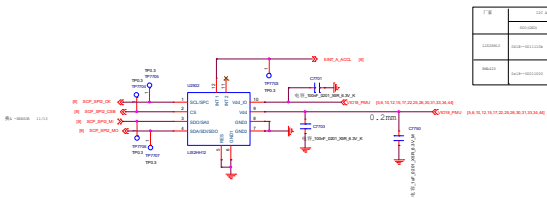
M-Sensor

厂号	001 000000
物料名称	0ch
物料规格	2ch

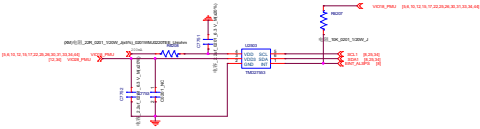


A sensor

default SPI (if sensor hub support)



ALS+PS



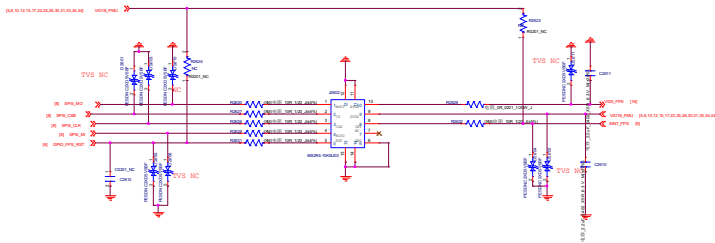
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 25.SENSOR

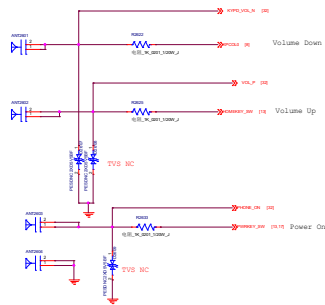
Size D	Project <Doc>	Rev V1
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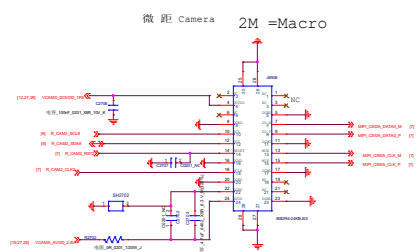
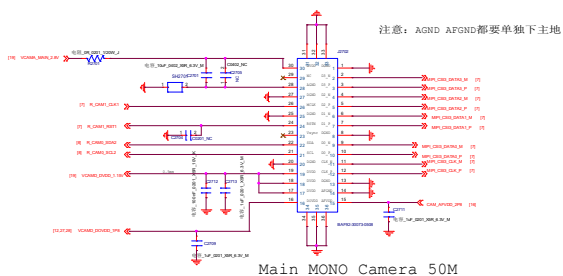
Finger Print



KEY_BT

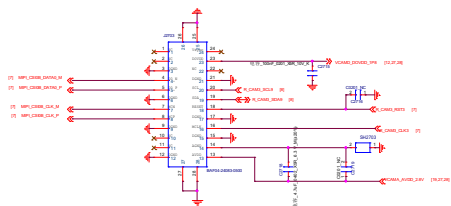


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		26.FP/KEY	
Size	Project	<Doc>	Rev
D			v1
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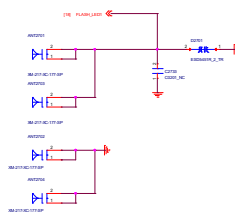


微距不贴

景深 Camera 2M=Depth



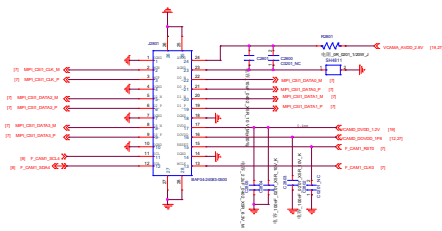
Flash LED 5V Boost



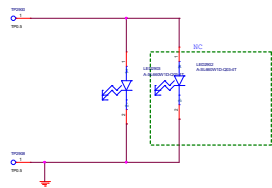
Flash 下方地要完整且足够大, 散热一定要好

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		27.REAR_CAM	
Size	Project	<Doc>	Rev
D			V1
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Front Camera - 8M

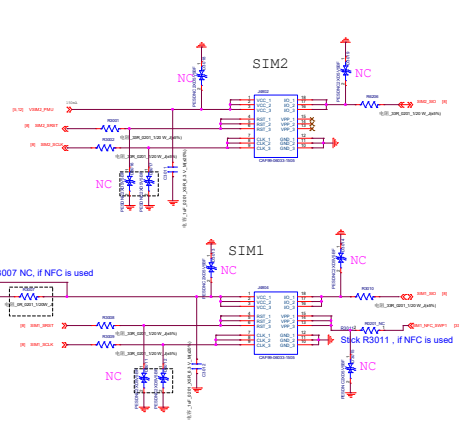
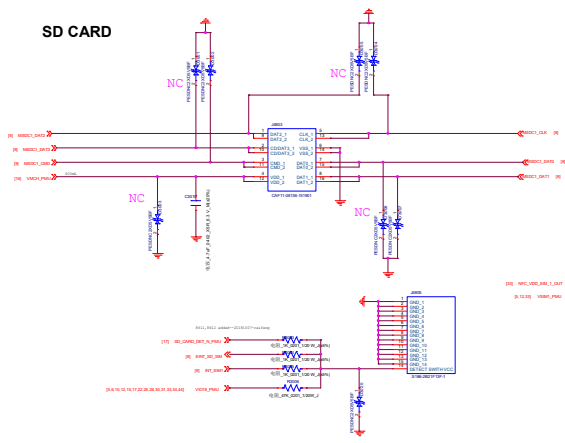


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		28.FRONT_CAM	
Size	Project	<Doc>	Rev
D			V1
Date: Monday, October 18, 2021		Sheet 28 of 50	

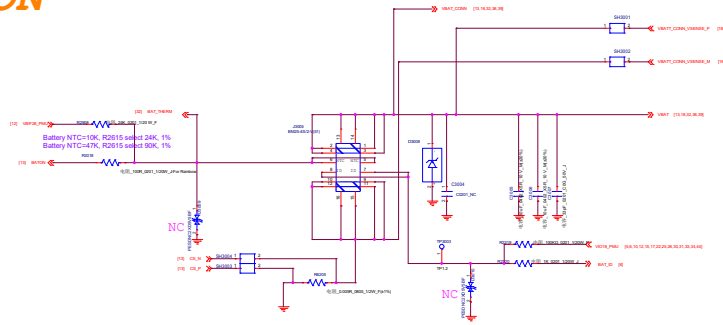


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		Sub_Charger_SMB1396	
Size	Project	<Doc>	Rev
D			V1
Date: Monday, October 18, 2021		Sheet 29 of	50

SD CARD

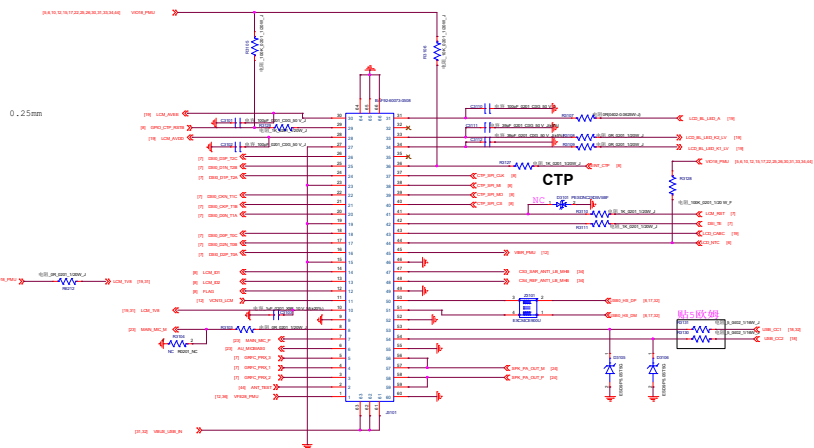


BATT CON

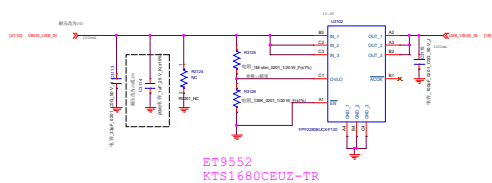


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		30.SIM/BATT	
Size	Project	<Doc>	Rev
D			V1
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LINK



OVP



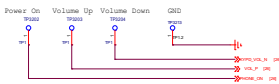
华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		31.60PIN_LCD_LINK/OVP	
Size D	Project	<Doc>	Rev V1
Date:	Friday, December 24, 2021	Sheet 31 of 50	

VBAT TEST POINT

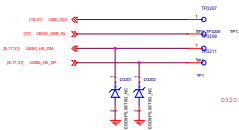


VBAT

KEY BTB



KEYPAD

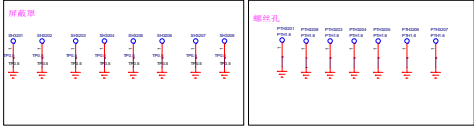


VCHG

UART TP

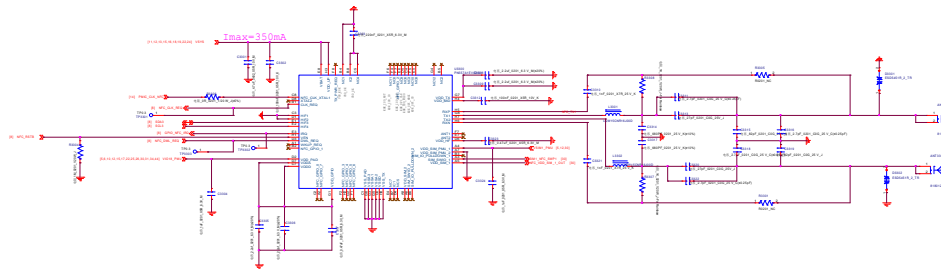


JTAG



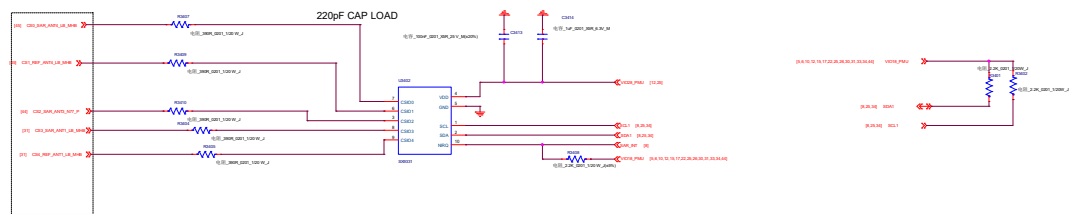
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 32.TEST_POINT/SHIELD		
Size D	Project <Doc>	Rev V1
Date: Monday, October 18, 2021	Sheet 32 of 50	

RF_NFC



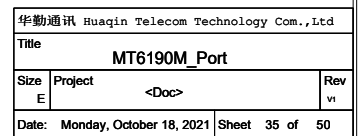
华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		RF_NFC	
Size	Project	<Doc>	Rev
E			v1
Date: Monday, October 18, 2021		Sheet 33 of	50

SAR_sensor



Design notice:
1) CS0_1 nets are differential pairs;
2) The width for each line, should be less than 0.1mm;
3) 0.2mm should be guaranteed between the ground and differential pairs;
4) And also, the distance between the differential line should be more than 0.2mm;
5) All the CSx nets should be protected by surrounding GND;

华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		SAR_sensor	
Size	Project	<Doc>	Rev
D			V1
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MT6190M PWR



华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title	MT6190M PWR
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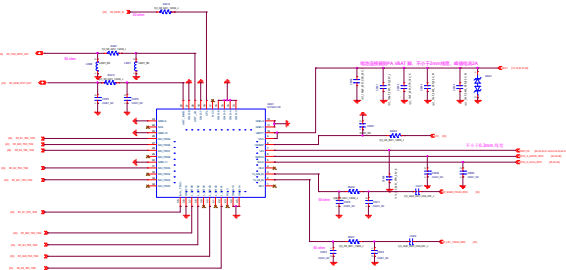
Size	Project	Rev
D	<Doc>	v1

Date: Monday, October 18, 2021 Sheet 36 of 50

华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title NC_FBRX		
Size E	Project <Doc>	Rev v1
Date: Monday, October 18, 2021	Sheet 37 of	50

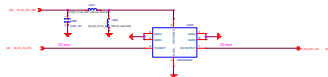
MAIN_ANT TXM_GSM

LB
MHB

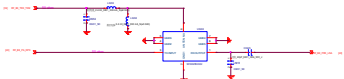


TRX_LB

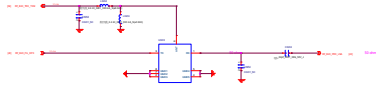
B5 TX



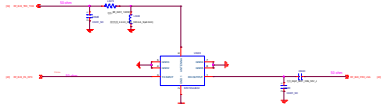
B8 TX



B20 TX

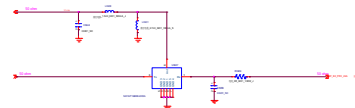


B28 TX

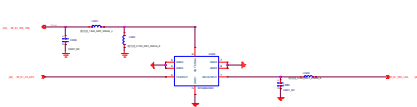


TRX_MHB

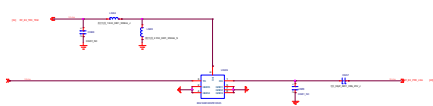
B2 TX



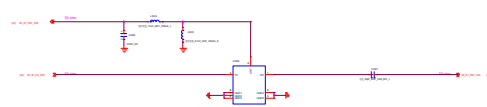
B1 TX



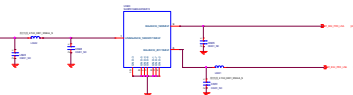
B3 TX



B7 TX

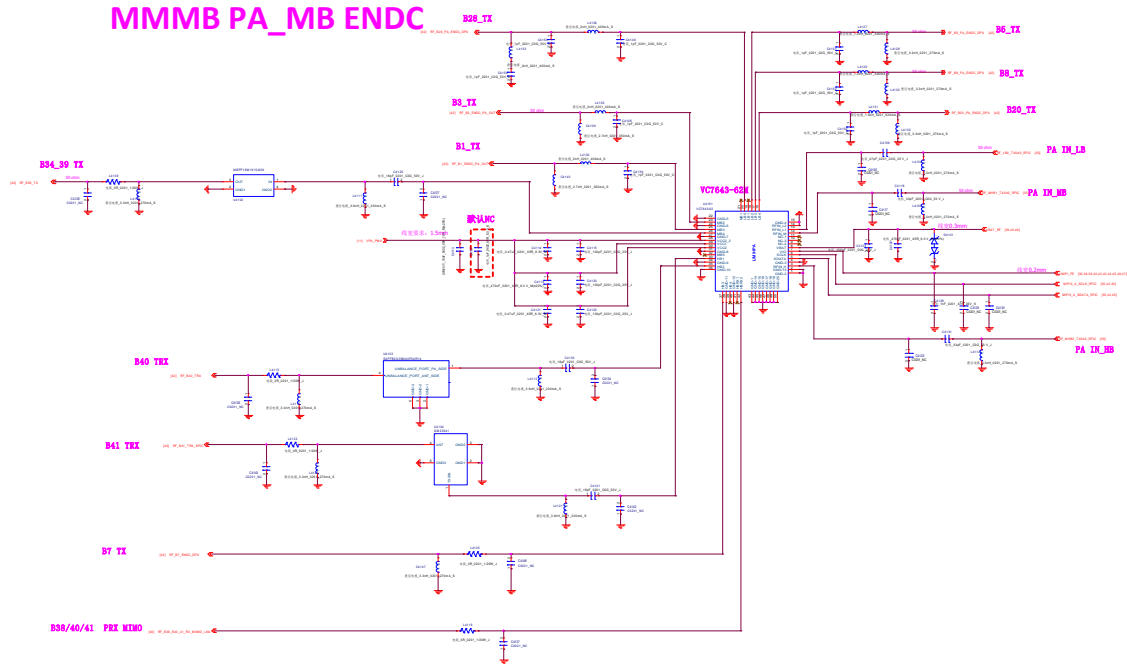


B34 /B39 PRX



华勤通讯 Huaqin Telecom Technology Co., Ltd			
Title TX_Module			
Size	Project	<Doc>	Rev
Custom			v1
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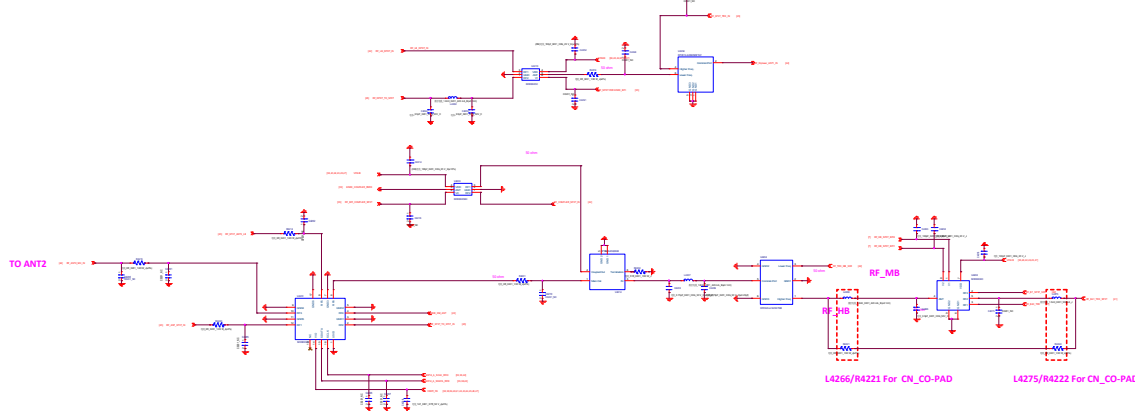
MMMB PA_MB ENDC



华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title MMBB PA_MB ENDC		
Size E	Project <Doc>	Rev v1
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D

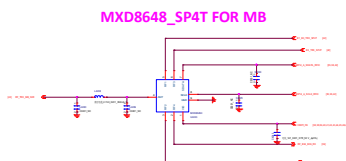
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C

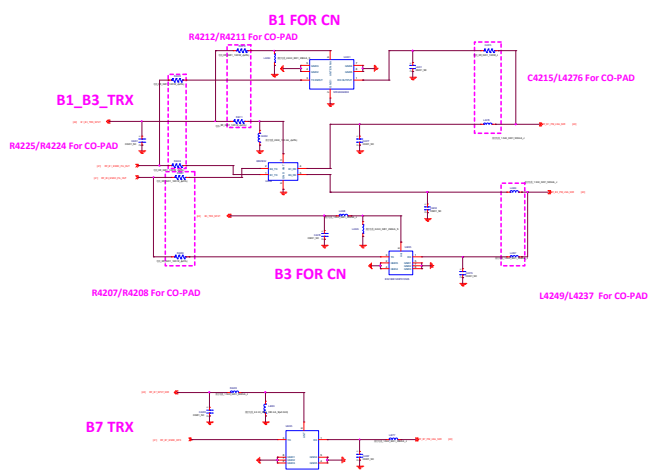
C

MHB



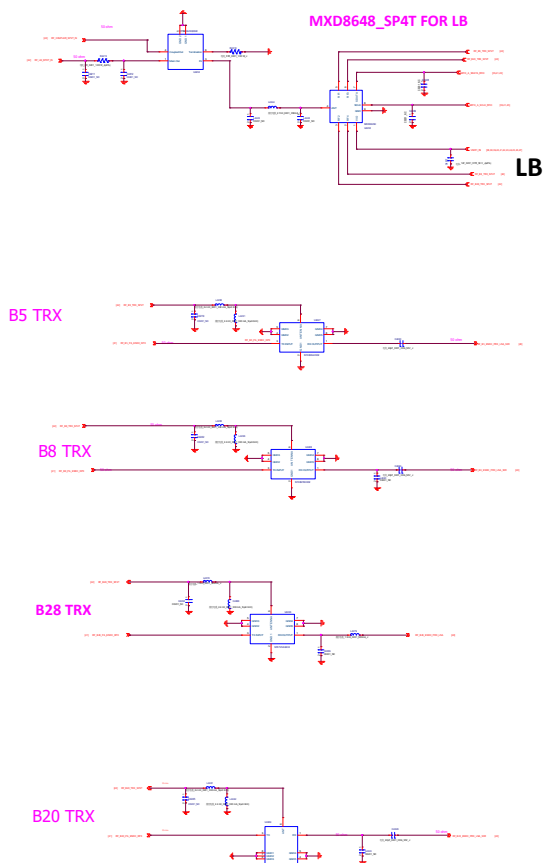
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B



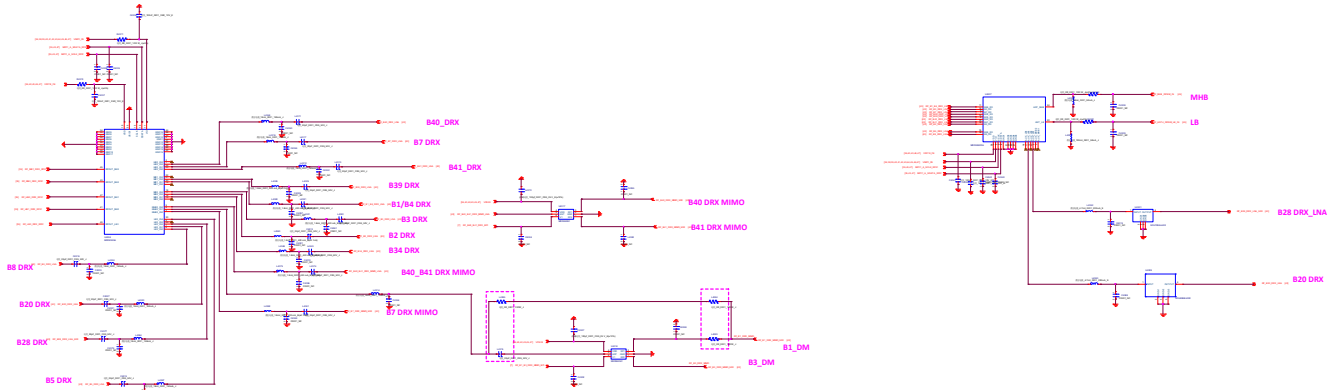
A

A

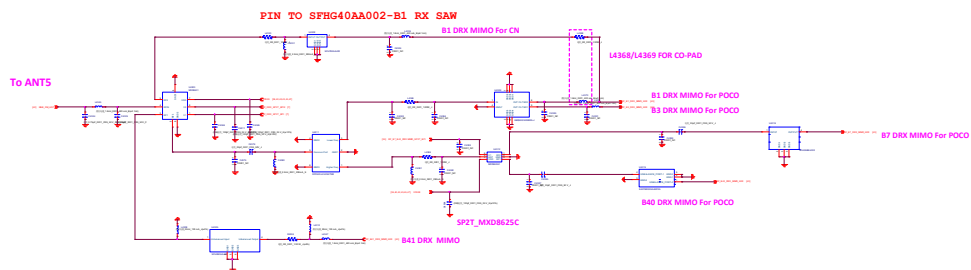


华为通信 Huagun Telecom Technology Co., Ltd			
Title: MMMB PA_VC7643_ENDC_MHB_TRX			
Size	Project	<Doc>	Rev
Custom			v1
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DRX_MXD98E2DA+VC3725

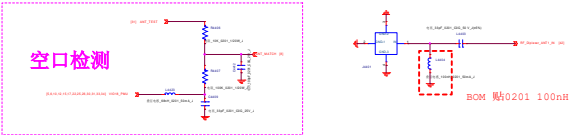


DRX_MIMO

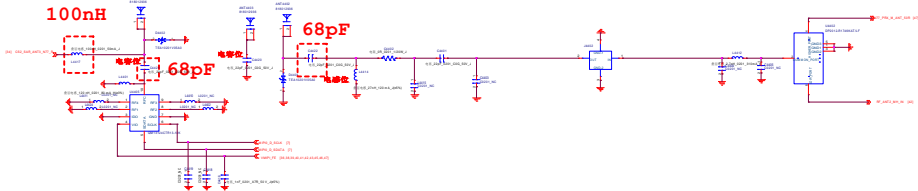


华勤通讯 Huaqin Telecom Technology Co., Ltd			
Title MXD98E2DA_VC3725			
Size	Project	Rev	
Custom	<Doc>	1	
Date: Friday, October 22, 2021	Sheet	43 of 50	

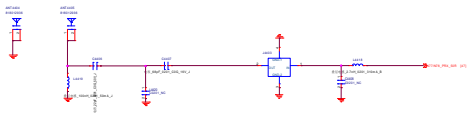
ANT1 (LB_MHB_P) to 小板



ANT2 (MHB_PM/N77_PRX_MIMO)

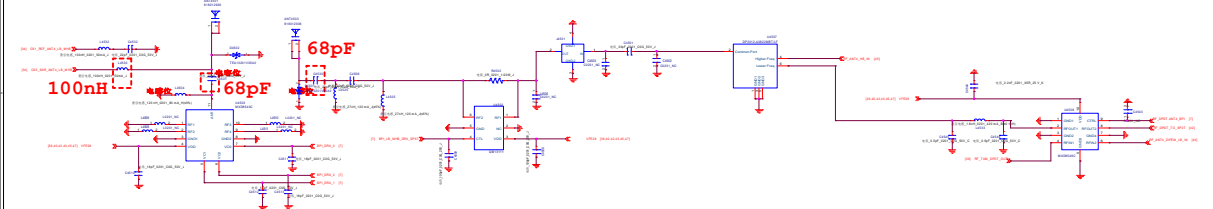


ANT3 (N77_PRX)

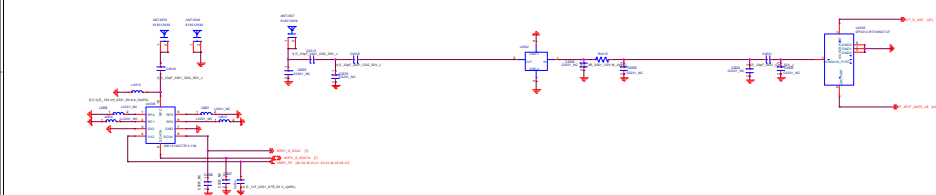


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		ANT_P_PM	
Size	Project	<Doc>	Rev
E			v1
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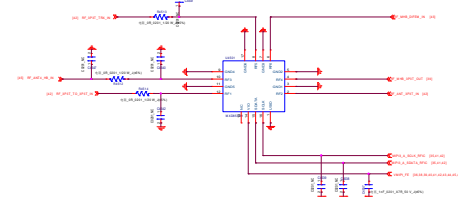
ANT4 (LB_MHB_D)



ANT5 (MHB_DM/N77_D)



AsDIV_3P3T



接地弹片

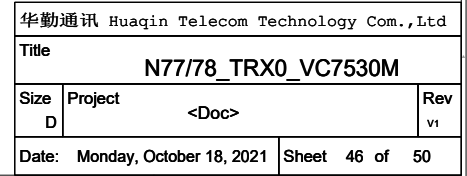


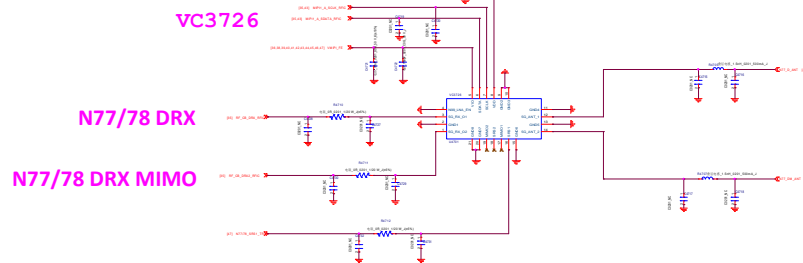
ANT6 (N77_DM_HB_DRX) to 小板



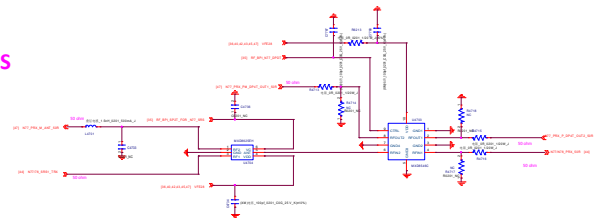
华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		ANT_D_DM	
Size	Project	<Doc>	Rev
E			v1
Date:	Friday, October 22, 2021	Sheet 45 of	50

NTC温度检测
见P10的R1002 and NTC1002

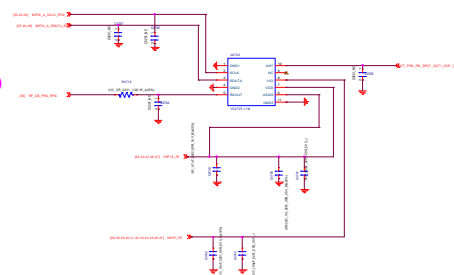




DPDT+SP2T FOR N77/78 SRS



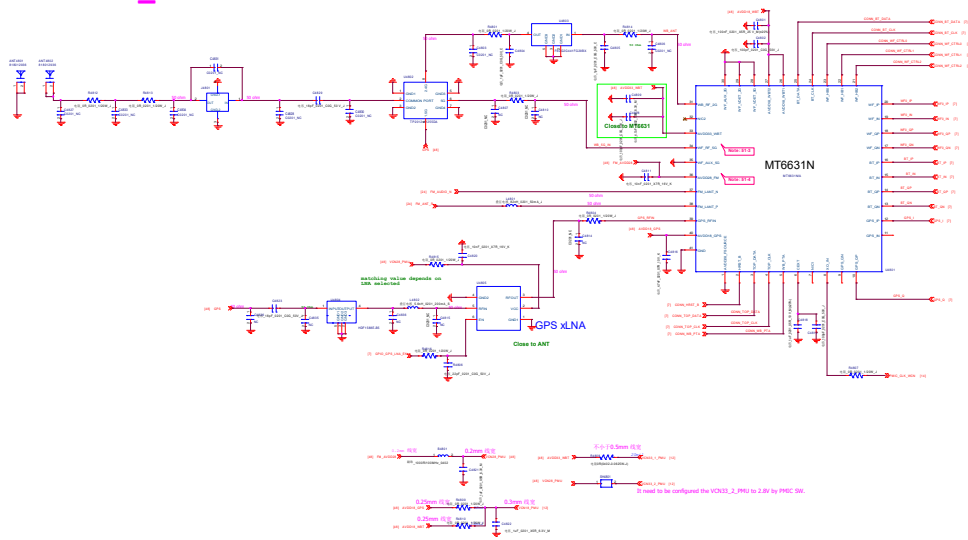
VC3727-11M_N77/78 PRX MIMO



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		VC3726	
Size	Project	<Doc>	Rev
E			v1
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ANT7 WiFi 2.4G & 5G_GPS

MT6631



Schematic design notice of "61_CONNECTIVITY_CONSYS_MT6631"

Note 51-1: For P6015 case, please select Q402 case or target one

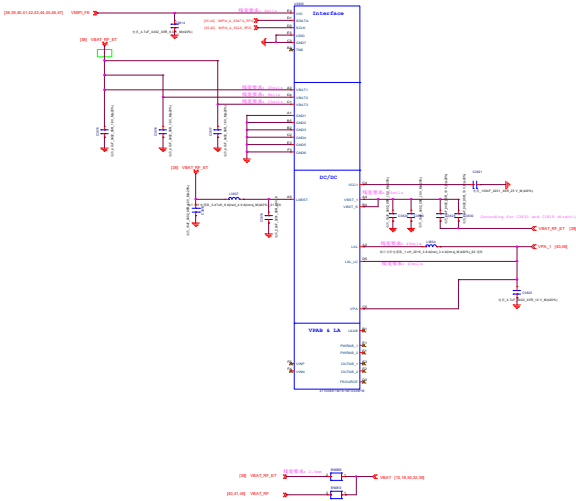
Note 51-2: Please refer to MT6631 Baseband design notice for VCN33_L2O selection guide

Note 51-3: If RFP1.55 not support, connect pin 34(RFP_RF_35) to GND

Note 51-4: Pin 36 (H40020_RF0) must be connected to VCN33 when IFTM not support

华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		MT6631	
Size	Project	<Doc>	Rev v1
Date:	Monday, October 18, 2021	Sheet 48 of 50	

MT6308 FOR Phase 5N PA_N77_PA



华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		RF_APT_MT6308M	
Size	Project	<Doc>	Rev
E			v1
Date: Monday, October 18, 2021		Sheet 50 of 50	